BEING AN IMMERSION IS FPPF LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites items from Spaces, Morphisms of Spaces, Topologies on Spaces and Descent plus same-chapter descent lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Definition of a property fppf local on the base. Stacks Project, Descent on Spaces, Definition definition-property-morphisms-local. Supplies the criterion structure the proof verifies.
- Base change of immersions. Stacks Project, Spaces, Lemma spaces-lemma-base-change-immersions. Proves the direct direction.
- Immersions are separated and locally quasi-finite. Stacks Project, Morphisms of Spaces, Lemmas spaces-morphisms-lemma-immersions-monomorphisms and spaces-morphisms-lemma-immersion-quasi-finite. Extracts the two descendable properties from each f_i.
- fpqc descent of separatedness and local quasi-finiteness. Stacks Project, Descent on Spaces, Lemmas lemma-descending-property-separated and lemma-descending-property-quasi-finite. Descends the two properties to f.
- Separated locally quasi-finite morphisms are representable. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-quasi-finite-separated-representable. Pivot: makes f representable.
- Testing representable morphisms on schemes. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-closed-immersion-local. Reduces to showing $Z \times_Y X \to Z$ is an immersion for every scheme $Z \to Y$.
- Refining fppf coverings by schemes. Stacks Project, Topologies on Spaces, Lemma spaces-topologies-lemma-refine-fppf-schemes. Produces the fppf covering of Z by schemes refining the pullback covering.
- fppf descent of immersions for schemes. Stacks Project, Descent, Lemma descent-lemma-descending-fppf-property-immersion. Concludes for $Z \times_Y X \to Z$.

Source reference. Original reference 02YW: sources/stacks-chow/tags.txt, source lines 3891–3891. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03YH: sources/stacks-chow/tags.txt, source lines 5184–5184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 042R: sources/stacks-chow/tags.txt, source lines 5338–5338. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03XM: sources/stacks-chow/tags.txt, source lines 5153–5153. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0421: sources/stacks-chow/tags.txt, source lines 5312–5312. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0427: sources/stacks-chow/tags.txt, source lines 5318–5318. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0418: sources/stacks-chow/tags.txt, source lines 5283–5283. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03M4: sources/stacks-chow/tags.txt, source lines 4736–4736. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 042T: sources/stacks-chow/tags.txt, source lines 5340–5340. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02YM: sources/stacks-chow/tags.txt, source lines 3881–3881. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is an immersion" is fppf local on the base.*

Proof. Let $f : X \to Y$ be a morphism of algebraic spaces. Let $\{Y_i \to Y\}_{i \in I}$ be an fppf covering of Y. Let $f_i : X_i \to Y_i$ be the base change of f.

If f is an immersion, then each f_i is an immersion by Spaces, Lemma 02YW. This proves the direct implication in Definition 03YH.

Conversely, assume each f_i is an immersion. By Morphisms of Spaces, Lemma 042R this implies each f_i is separated. By Morphisms of Spaces, Lemma 03XM this implies each f_i is locally quasi-finite. Hence we see that f is locally quasi-finite and separated, by applying Lemmas 0421 and 0427. By Morphisms of Spaces, Lemma 0418 this implies that f is representable!

By Morphisms of Spaces, Lemma 03M4 it suffices to show that for every scheme Z and morphism $Z \to Y$ the base change $Z \times_Y X \to Z$ is an immersion. By Topologies on Spaces, Lemma 042T we can find an fppf covering $\{Z_i \to Z\}$ by schemes which refines the pullback of the covering $\{Y_i \to Y\}$ to Z. Hence we see that $Z \times_Y X \to Z$ (which is a morphism of schemes according to the result of the preceding paragraph) becomes an immersion after pulling back to the members of an fppf (by schemes) of Z. Hence $Z \times_Y X \to Z$ is an immersion by the result for schemes, see Descent, Lemma 02YM. □